CP2 Histology: top 50 questions

Topic 9A · Peripheral blood (H11A)

Centrifuged blood, hematocrit, plasma vs serum

1. *2026 TA review*
A centrifuged blood specimen demonstrates a hematocrit of 45% and a negligible buffy coat. Approximately what percentage of the blood volume is plasma?

A. 25%
B. 35%
C. 45%
D. 55%
E. 75%

Answer: D. 55%

Plasma = 100% minus hematocrit (45%) minus a negligible buffy coat = **55%**.
C (45%): that is the red cell layer, not plasma.

2. *2026 TA review*
A tube of blood is allowed to clot and is then centrifuged. The supernatant fluid obtained is best described as:

A. Plasma containing coagulation factors
B. Serum lacking coagulation factors consumed during clotting
C. Whole blood without erythrocytes
D. Buffy coat fluid
E. Lymph

Answer: B. Serum lacking coagulation factors consumed during clotting

Blood that is **allowed to clot** gives **serum**: plasma minus the fibrinogen and factors used up in the clot.
A (plasma): plasma comes from **anticoagulated** blood and keeps its clotting factors.
D (buffy coat): a layer of leukocytes and platelets, not a fluid.
E (lymph): tissue fluid in lymphatic vessels, not from a tube of blood.

Erythrocytes

3. *2026 TA review*
A patient has erythrocytes with reduced membrane flexibility, making it difficult for them to pass through narrow capillaries. Which structural component normally supports the biconcave shape and flexibility of erythrocytes?

A. Myeloperoxidase in primary granules
B. Spectrin lattice anchored by ankyrin
C. Hemoglobin bound to fibrinogen
D. Marginal microtubule bundle
E. Histamine-containing granules

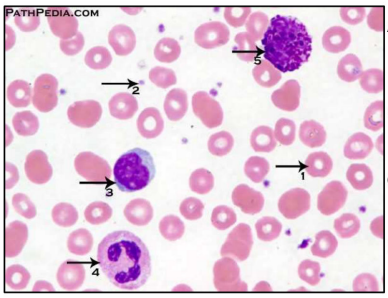
Answer: B. Spectrin lattice anchored by ankyrin

Red cell shape and flexibility come from the **spectrin lattice anchored by ankyrin** under the membrane.
A: myeloperoxidase is in **neutrophil** azurophilic granules.
D (marginal microtubule bundle): keeps the disc shape of **platelets**.
E: histamine granules belong to basophils and mast cells.

Leukocyte emigration and classification

4. *2026 blood session*
Identify the labeled formed elements in this peripheral smear

Q1. Identify the labeled formed elements in this peripheral smear



A. 1)RBC, 2)extruded nucleus, 3)lymphocyte, 4)eosinophil, 5)basophil

B. 1)RBC, 2)reticulocyte, 3)monocyte, 4)neutrophil, 5) basophil

C. 1)RBC, 2)platelet, 3)lymphocyte, 4)neutrophil, 5)basophil

D. 1)Reticulocyte, 2) platelet, 3)lymphocyte, 4)neutrophil, 5)basophil

A. 1)RBC, 2)extruded nucleus, 3)lymphocyte, 4)eosinophil, 5)basophil

B. 1)RBC, 2)reticulocyte, 3)monocyte, 4)neutrophil, 5) basophil

C. 1)RBC, 2)platelet, 3)lymphocyte, 4)neutrophil, 5)basophil

D. 1)Reticulocyte, 2) platelet, 3)lymphocyte, 4)neutrophil, 5)basophil

Answer: C. 1)RBC, 2)platelet, 3)lymphocyte, 4)neutrophil, 5)basophil

1 = pink disc, pale center, no nucleus: **RBC**. **2** = tiny purple fragment: **platelet**. **3** = round dense nucleus, thin blue rim: **lymphocyte**. **4** = 3-5 lobed nucleus, pale lilac cytoplasm: **neutrophil**. **5** = dark blue-black granules hiding the nucleus: **basophil**.
A: 2 is far too small for a nucleus; 4 lacks red-orange granules (not eosinophil).
B: a reticulocyte is RBC-sized; 3 has too little cytoplasm for a monocyte.

Neutrophils

5. *2026 Canvas practice quiz*

Match each item to the correct description — First leukocyte to arrive at the site of infection, if the patient is female a barr body will be visible

A. Basophil
B. Eosinophil
C. Monocyte
D. Platelet
E. Lymphocyte
F. Neutrophil

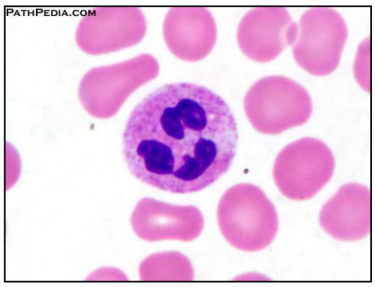
Answer: F. Neutrophil

First leukocyte to an infection plus a **Barr body** in females = **neutrophil**.
B (eosinophil): parasites, allergy.
C (monocyte): arrives later, becomes a macrophage.
E (lymphocyte): adaptive immunity, no lobed nucleus.

6. *2026 blood session*

Which of the following is true of this cell?

Q2. Which of the following is true of this cell?



A. It is a band neutrophil
B. It is the second most abundant leukocyte
C. It is not a granulocyte
D. It is a phagocytic cell
E. It does not contain azurophilic granules

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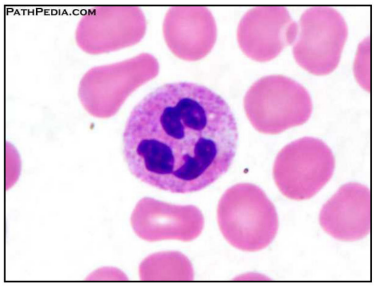
Answer: D. It is a phagocytic cell

This is a **segmented neutrophil** (3-5 lobes, pale lilac cytoplasm, about twice an RBC). Neutrophils are **active phagocytes** of bacteria.
A: a band cell has a curved, **unsegmented** horseshoe nucleus.
B: it is the **most** abundant leukocyte (50-70%).
C: it is a granulocyte.
E: it has both azurophilic (primary) and specific granules.

7. *2026 blood session*

Which of the following is true of this cell?

Q2. Which of the following is true of this cell?



A. It is a band neutrophil
B. It is the second most abundant leukocyte
C. Transendothelial migration is facilitated by the elongated and segmented nucleus
D. It is a weakly phagocytic cell
E. It contains azurophilic, but not specific granules

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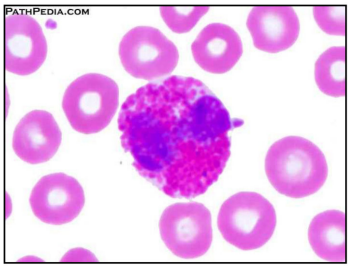
Answer: C. Transendothelial migration is facilitated by the elongated and segmented nucleus

A **segmented neutrophil**: its thin, **elongated, lobed nucleus** lets it squeeze between endothelial cells during **transendothelial migration** (diapedesis).
A: a band cell has an unsegmented horseshoe nucleus; this one has separate lobes.
B: it is the **most** abundant leukocyte.
D: it is an **avid** phagocyte.
E: it has **both** azurophilic and specific granules.

Eosinophils and basophils

8. *2026 blood session*
Which of the following is true of this cell?

Q3. Which of the following is true of this cell?



A. Its main function is to destroy viruses
B. It is a band form
C. It is not involved in allergic reactions
D. The granules contain major basic protein
E. It is almost as abundant as neutrophils

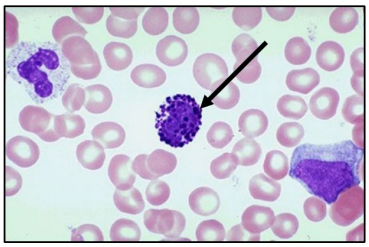
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C. It is not involved in allergic reactions
D. The granules contain major basic protein
E. It is almost as abundant as neutrophils

Answer: D. The granules contain major basic protein

This is an **eosinophil**: bilobed nucleus, large red-orange granules. Its specific granules hold **MBP (major basic protein)**, plus eosinophil peroxidase and histaminase.
A: killing virus-infected cells is the job of lymphocytes (NK = natural killer cells, cytotoxic T cells).
B: a band has an unsegmented horseshoe nucleus and no red granules.
C: eosinophils do take part in allergy.
E: eosinophils are only 1–4% of WBCs (white blood cells); neutrophils are 50–70%.

9. *2025 faculty blood session*
Q5: What are the biologically active molecules contained within the granules of the cell indicated by an arrow?

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A. Histamine and DNase
B. Histamine and ribosomal RNA
C. Histamine and heparin
D. Lysosomes and hydrogen peroxide
E. Calcium and lipase

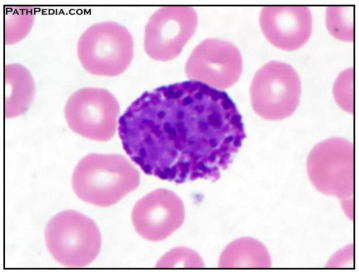
A. Histamine and DNase
B. Histamine and ribosomal RNA
C. Histamine and heparin
D. Lysosomes and hydrogen peroxide
E. Calcium and lipase

Answer: C. Histamine and heparin

The arrow is on a small cell whose nucleus is hidden by **coarse dark blue-black granules**: a **basophil**. Its granules hold **histamine and heparin** (plus eosinophil chemotactic factor).
A, B, E: DNase, ribosomal RNA, calcium and lipase are not basophil mediators.
D (lysosomes, hydrogen peroxide): neutrophil killing machinery; the neutrophil is the cell at the left edge.

10. *2026 blood session*
Which of the following is true of this cell?

Q4. Which of the following is true of this cell?



A. Its main function is phagocytosis
B. Its main purpose is to respond to produce antibodies
C. It performs functions similar to the mast cell
D. Its granules do not contain histamine
E. Its nucleus is not lobulated

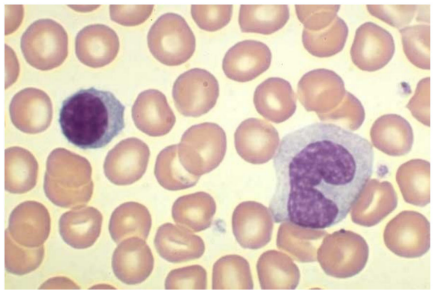
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B. Its main purpose is to respond to produce antibodies
C. It performs functions similar to the mast cell
D. Its granules do not contain histamine
E. Its nucleus is not lobulated

Answer: C. It performs functions similar to the mast cell

This is a **basophil**: dark purple granules hide the nucleus. It has IgE (immunoglobulin E) receptors and granules of **histamine and heparin**, so it works like a **mast cell** in allergy.
A: phagocytosis is the main job of neutrophils and monocytes/ macrophages.
B: antibodies come from plasma cells.
D: basophil granules do contain histamine.
E: the nucleus is bilobed or S-shaped, just hidden by granules.

Lymphocytes and monocytes

11. *2026 blood lab deck*
The leukocyte on the left is a…

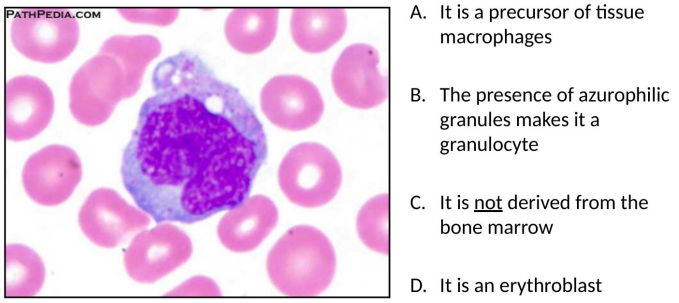


A. Neutrophil
B. Eosinophil
C. Basophil
D. Monocyte
E. Lymphocyte

Answer: E. Lymphocyte

Small lymphocyte: about the **size of an RBC** (red blood cell), a **round, dense, dark nucleus** filling almost the whole cell, and only a **thin rim of pale blue cytoplasm**.
Monocyte: much larger, indented kidney-shaped nucleus, plenty of grey-blue cytoplasm.
Basophil: dark granules hide the nucleus.
D (monocyte): that is the larger cell on the right.

12. *2026 blood session*
Which of the following is true of this cell?



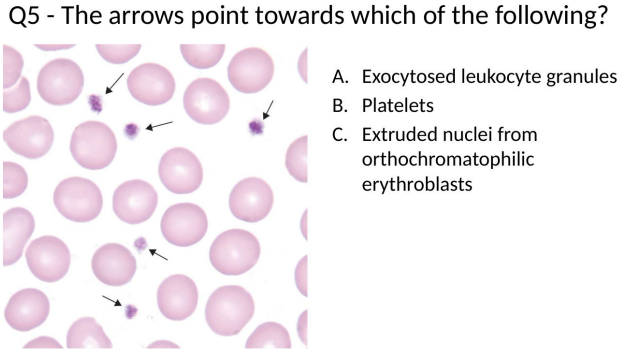
A. It is a precursor of tissue macrophages
B. The presence of azurophilic granules makes it a granulocyte
C. It is not derived from the bone marrow
D. It is an erythroblast

Answer: A. It is a precursor of tissue macrophages

Monocyte: large cell, indented nucleus with lacy chromatin, grey-blue cytoplasm with small vacuoles. It leaves the blood to become a **tissue macrophage**.
B: its azurophilic granules are just lysosomes; with no specific granules it is an **agranulocyte**.
C: monocytes are made in the bone marrow.
D: erythroblasts live in marrow and have a round, non-indented nucleus.

Platelet structure: LM and EM

13. *2026 blood lab deck*
The arrows point towards which of the following?



A. Exocytosed leukocyte granules
B. Platelets
C. Extruded nuclei from orthochromatophilic erythroblasts

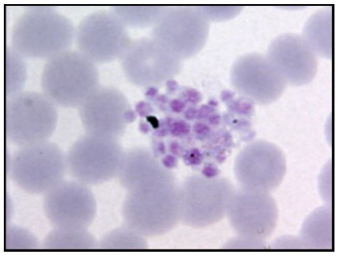
Answer: B. Platelets

Platelets: tiny (2–3 µm) **anucleate** purple fragments, often in **clumps**, each with a pale rim (**hyalomere**) and dark granular center (**granulomere**).
Extruded nuclei of orthochromatophilic erythroblasts are eaten by marrow macrophages and are not seen in normal blood.
Leukocyte granules are not found free in a smear.

Platelet origin and hemostasis

14. *2025 faculty blood session*
Which of the following is true of the central small structures in this blood smear?

Q9. Which of the following is true of the central small structures in this blood smear?



A. Electron microscopy shows them to be perfectly spherical
B. They have a peripheral granulomere and a central hyalomere
C. They have a minute central nucleus
D. They are agranular
E. They are derived from polyploid bone marrow cells

A. Electron microscopy shows them to be perfectly spherical
B. They have a peripheral granulomere and a central hyalomere
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E. They are derived from polyploid bone marrow cells

Answer: E. They are derived from polyploid bone marrow cells

The clump of tiny fragments is **platelets**; they are shed from **megakaryocytes**, giant **polyploid** bone marrow cells.
A: they are discs, not spheres.
B: reversed; hyalomere is peripheral, granulomere central.
C: platelets have no nucleus.
D: they have alpha and delta granules.

Topic 9B · Hematopoiesis & bone marrow (H11B)

Hematopoietic sites by age; marrow conversion

15. *2026 TA review*
Question: A bone marrow biopsy shows hematopoietic tissue replaced by adipose tissue in most long bones. This finding represents:

A. Normal age-related replacement of red marrow by yellow marrow
B. Pathologic aplastic anemia
C. Myelophthisic anemia
D. Bone marrow fibrosis
E. Iron deficiency anemia

Answer: A. Normal age-related replacement of red marrow by yellow marrow

Red marrow is slowly replaced by fat (**yellow marrow**) with age. By adulthood, long-bone shafts are mostly fatty and red marrow stays in the axial skeleton. This is **normal**.
B (aplastic anemia): fatty, empty marrow in the axial skeleton too, with pancytopenia (all blood counts low); not the normal long-bone pattern.
C (myelophthisic) / D (fibrosis): marrow replaced by tumor or fibrous tissue, not fat.

Red marrow histology, megakaryocytes, sinusoids (incl. IDs)

16. *2026 TA review*

How do bone marrow cells enter circulation?

A. Passing through the discontinuous sinusoidal endothelium

B. Passing through the fenestrated endothelium

C. Passing through the continuous endothelium

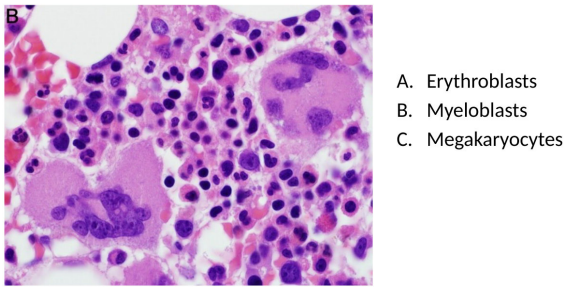
D. Passing through the tight junctions

Answer: A. Passing through the discontinuous sinusoidal endothelium

New blood cells reach the blood by squeezing through the **discontinuous endothelium of marrow sinusoids**. The sinusoids have gaps and a broken basal lamina.
B (fenestrated): small pores with diaphragms (kidney, endocrine glands); too small for whole cells.
C (continuous): no gaps (muscle, lung, brain).
D (tight junctions): seal cells together; cells don't pass through them.

17. *2026 blood lab deck*

In this normal bone marrow smear, the large cells with lobulated nuclei are…



A. Erythroblasts

B. Myeloblasts

C. Megakaryocytes

Answer: C. Megakaryocytes

The two **giant cells** with broad pink granular cytoplasm and one multilobed nucleus are **megakaryocytes**, the largest marrow cells.
A (erythroblasts): small cells with round, very dark nuclei; many sit around them.
B (myeloblasts): medium cells with one round nucleus, fine chromatin and nucleoli; no lobes.

Erythropoiesis stages and reticulocytes (incl. IDs)

18. *2026 blood session · 2025 faculty blood session*

In order of development, what are the 6 stages of erythropoiesis?

A. Proerythroblast, basophilic erythroblast, polychromatophilic erythroblast, orthochromatophilic erythroblast, reticulocyte, erythrocyte

B. Erythroblast, basophilic erythroblast, polychromatophilic erythroblast, orthochromatophilic erythroblast, reticulocyte, erythrocyte

C. Proerythroblast, orthochromatophilic erythroblast, basophilic erythroblast, polychromatophilic erythroblast, reticulocyte, erythrocyte

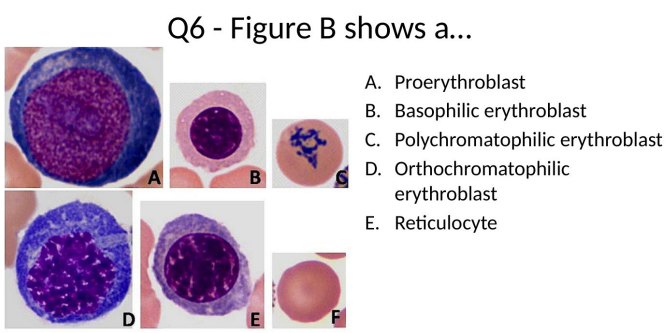
D. Proerythroblast, polychromatophilic erythroblast, basophilic erythroblast, orthochromatophilic erythroblast, reticulocyte, erythrocyte

Answer: A. Proerythroblast, basophilic erythroblast, polychromatophilic erythroblast, orthochromatophilic erythroblast, reticulocyte, erythrocyte

Erythropoiesis: **proerythroblast → basophilic → polychromatophilic → orthochromatophilic erythroblast → reticulocyte → erythrocyte**.
Trend: cell shrinks, nucleus condenses then is expelled, cytoplasm goes **blue (ribosomes) → mixed → pink (hemoglobin)**.
B: 'erythroblast' is not the first stage; it begins with the **pro**erythroblast.
C, D: put the color stages out of order; blue (basophilic) → mixed (polychromatophilic) → pink (orthochromatophilic).

19. *2026 blood lab deck*

Figure B shows a…



A. Proerythroblast

B. Basophilic erythroblast

C. Polychromatophilic erythroblast

D. Orthochromatophilic erythroblast

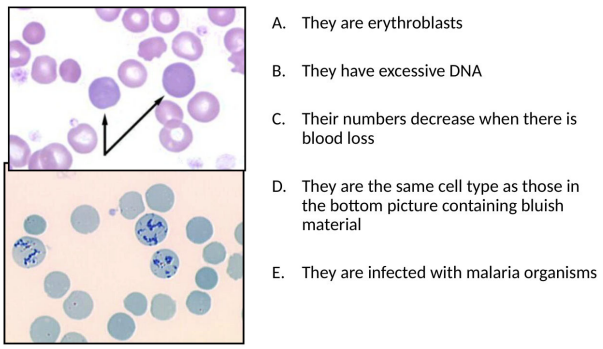
E. Reticulocyte

Answer: D. Orthochromatophilic erythroblast

D: Figure B is small, with **pink (hemoglobin-filled) cytoplasm** and a **small, dense, round nucleus**: orthochromatophilic erythroblast, the last nucleated stage.
A (proerythroblast, Figure A): largest cell, fine chromatin, nucleoli, deep-blue rim.
B (basophilic, Figure D): large, intensely blue cytoplasm, coarser chromatin.
C (polychromatophilic, Figure E): grey-lilac (mixed blue + pink) cytoplasm, checkerboard nucleus.
E (reticulocyte, Figure C): no nucleus.

20. *2026 blood session*

Which of the following is true of the cells at the tips of the arrows?



A. They are erythroblasts

B. They have excessive DNA

C. Their numbers decrease when there is blood loss

D. They are the same cell type as those in the bottom picture containing bluish material

E. They are infected with malaria organisms

Answer: D. They are the same cell type as those in the bottom picture containing bluish material

D: the arrowed cells are **reticulocytes** on Wright stain: anucleate, **slightly larger** than RBCs (red blood cells), **bluish-lilac**. The bottom (new methylene blue) cells with a **blue RNA network** are the same cells.
A (erythroblasts): would have a nucleus; these have none.
B (excess DNA): no nucleus, so no DNA; the blue tint is residual RNA.
C: blood loss makes the marrow release **more** reticulocytes, not fewer.
E (malaria): the blue material is precipitated ribosomal RNA, not parasites.

Granulopoiesis; erythropoiesis vs granulopoiesis (incl. IDs)

21. *2026 blood session*
In order of development, what are the 6 stages of neutrophil granulopoiesis?

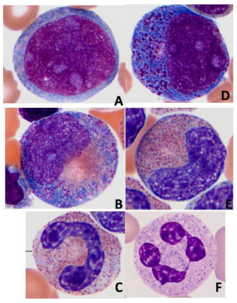
- **A.** Myeloblast, promyelocyte, myelocyte, metamyelocyte, band, neutrophil
- **B.** Promyelocyte, myeloblast, myelocyte, metamyelocyte, band, neutrophil
- **C.** Myelocyte, myeloblast, promyelocyte, metamyelocyte, band, neutrophil
- **D.** Metamyelocyte, myeloblast, promyelocyte, myelocyte, band, neutrophil
- **E.** Band, myeloblast, promyelocyte, myelocyte, metamyelocyte, neutrophil

Answer: A. Myeloblast, promyelocyte, myelocyte, metamyelocyte, band, neutrophil

Granulopoiesis: **myeloblast → promyelocyte → myelocyte → metamyelocyte → band → segmented neutrophil**.
Clues: azurophilic granules appear at the promyelocyte, **specific granules at the myelocyte**, the nucleus then indents (kidney), becomes a horseshoe (band) and finally segments.
B–E: each starts with a later cell; the **blast** (no granules) always comes first.

22. *2026 blood lab deck*
Figure A shows a…

Q8 - Figure A shows a...



A. Myeloblast
B. Promyelocyte
C. Neutrophilic myelocyte
D. Neutrophilic metamyelocyte
E. Neutrophilic band

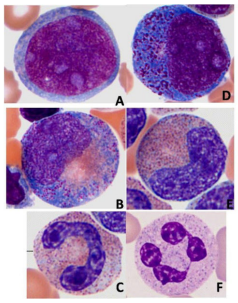
- **A.** Myeloblast
- **B.** Promyelocyte
- **C.** Neutrophilic myelocyte
- **D.** Neutrophilic metamyelocyte
- **E.** Neutrophilic band

Answer: A. Myeloblast

A: Figure A is large with a **round nucleus of fine chromatin and several nucleoli** and a rim of blue cytoplasm with **no granules** = myeloblast.
B (promyelocyte, Figure D): same size but cytoplasm full of dark **azurophilic granules**.
C (myelocyte, Figure B): round-oval nucleus, pale **specific granules**, pale area by the nucleus.
D (metamyelocyte, Figure E): kidney-shaped nucleus.

23. *2026 blood lab deck*
Figure C shows a…

Q9 - Figure C shows a...



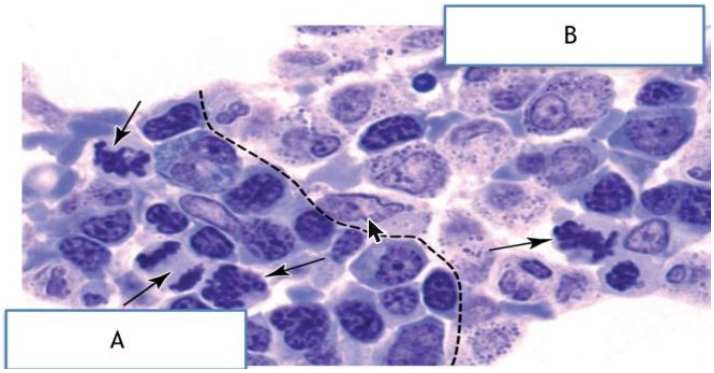
A. Myeloblast
B. Promyelocyte
C. Neutrophilic myelocyte
D. Neutrophilic metamyelocyte
E. Neutrophilic band

- **A.** Myeloblast
- **B.** Promyelocyte
- **C.** Neutrophilic myelocyte
- **D.** Neutrophilic metamyelocyte
- **E.** Neutrophilic band

Answer: E. Neutrophilic band

E: Figure C has a **horseshoe/C-shaped nucleus of even width**, not yet pinched into lobes = **band** neutrophil.
D (metamyelocyte, Figure E): nucleus only **indented** (kidney-shaped), still broad.
C (myelocyte, Figure B): round-oval nucleus.
Not the segmented neutrophil (Figure F): that one has 3–5 lobes joined by thin strands.

24. *2026 Canvas practice quiz*
In the image below we observe a bone marrow micrograph. "A" is labeling the side of the image where ____ is predominantly seen.



- **A.** thrombopoiesis
- **B.** erythropoiesis
- **C.** granulopoiesis

Answer: B. erythropoiesis

B: side A (lower left of the dashed line) has **darker, tightly packed cells with round, condensed nuclei** and **no granules** = **erythropoiesis**. The arrows mark mitoses.
C (granulopoiesis): that is side B, with paler cells, **visible cytoplasmic granules** and indented nuclei.
A (thrombopoiesis): would need giant multilobed **megakaryocytes**; none here.

Lab clinical correlates: anemia, inflammation

25. *2025 faculty lab deck*
Case No. 5: In an anemia workup, which of the following should be ordered first to test for effective erythropoiesis?

A. Bone marrow aspirate
B. Bone marrow trephine biopsy
C. Red cell survival study
D. Reticulocyte count
E. Serum bilirubin

Answer: D. Reticulocyte count

The **reticulocyte count** shows how fast the marrow is releasing new RBCs. It is the first, simplest test of effective erythropoiesis: high after blood loss or hemolysis, low in marrow failure.
A / B (marrow aspirate / trephine biopsy): invasive; done later if needed.
C (red cell survival study): measures RBC destruction, not production.
E (bilirubin): a marker of hemolysis.

26. *2026 TA review*
A 19 year old male presents to the emergency department with severe pain in his chest and extremities. He reports multiple similar episodes, particularly during periods of dehydration and physical exertion. A peripheral blood smear reveals numerous elongated, crescent-shaped erythrocytes. Which of the following mechanisms most directly explains the abnormal erythrocyte morphology?

A. Decreased production of hemoglobin due to iron deficiency
B. Polymerization of abnormal hemoglobin S under low-oxygen conditions
C. Autoimmune destruction of circulating erythrocytes
D. Failure of erythrocytes to extrude their nuclei during maturation
E. Increasing production of erythrocytes due to a JAK2 mutation

Answer: B. Polymerization of abnormal hemoglobin S under low-oxygen conditions

B: crescent RBCs (red blood cells) = **sickle cell anemia**. Mutant **hemoglobin S polymerizes when oxygen is low** (dehydration, exertion), bending the cell into a rigid sickle.
A (iron deficiency): gives small pale cells, not sickles.
C (autoimmune hemolysis): gives spherocytes.
D: failed nuclear extrusion would leave nucleated RBCs.
E (JAK2 = Janus kinase 2): more RBCs of normal shape.

Topic 10 · Lymphoid cells, tissues & organs (H14)

Lymphoid organs and lymph circulation

27. *2026 TA review*
Which statement best describes the defining functional distinction between primary (central) and secondary (peripheral) lymphoid organs?

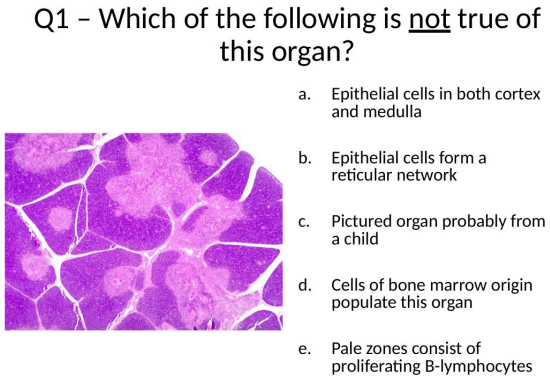
A. Primary organs filter lymph, while secondary organs filter blood.
B. Primary organs contain encapsulated nodules, while secondary organs are unencapsulated.
C. Primary organs are where lymphocytes form and mature, while secondary organs are where immune reactions take place.
D. Primary organs rely on high endothelial venules (HEVs) for cell entry, while secondary organs do not.

Answer: C. Primary organs are where lymphocytes form and mature, while secondary organs are where immune reactions take place.

Primary organs (marrow, thymus) = where lymphocytes **form and mature**; **secondary** organs (nodes, spleen, MALT) = where **immune reactions** happen.
A: nodes filter lymph and the spleen filters blood, and both are secondary.
B: the thymus has a capsule; MALT is unencapsulated; this is not the defining split.
D: HEVs (high endothelial venules) are in secondary organs like lymph nodes, not primary ones.

Thymus: architecture, age changes, identification

28. *2026 lymphoid session*
Which of the following is not true of this organ?

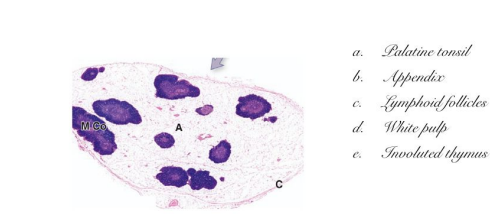


A. Epithelial cells in both cortex and medulla
B. Epithelial cells form a reticular network
C. Pictured organ probably from a child
D. Cells of bone marrow origin populate this organ
E. Pale zones consist of proliferating B-lymphocytes

Answer: E. Pale zones consist of proliferating B-lymphocytes

This is a **child's thymus**: lobules with dark cortex, pale medulla, no fat. The pale zones are **medulla** (T cells, TECs, Hassall corpuscles), **not B-cell germinal centers**, so E is false.
A, B: true. TECs (thymic epithelial cells) sit in both zones and form a cytoreticulum.
C: true. No fatty involution.
D: true. T-cell progenitors come from bone marrow.

29. *past faculty session*
Q5. - This image depicts which of the following?



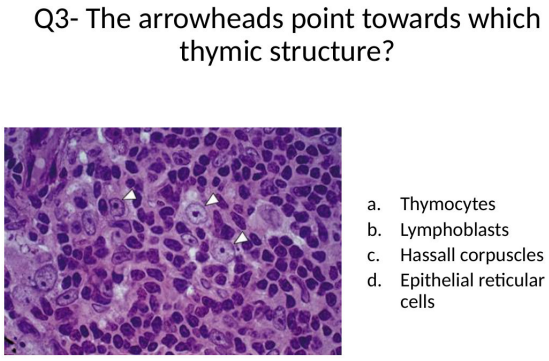
A. Palatine tonsil
B. Appendix
C. Lymphoid follicles
D. White pulp
E. Involuted thymus

Answer: E. Involuted thymus

Mostly **adipose tissue (A)** inside a thin capsule (C), with small islands of dark lymphoid tissue showing remnant **cortex and medulla (Co, M)** = **involuted adult thymus**.
A (palatine tonsil): would show stratified squamous epithelium and crypts.
B (appendix): would show gut wall: mucosa, glands, muscle.
C, D (lymphoid follicles, white pulp): not floating in fat; white pulp sits in red pulp around a central artery.

Thymic epithelial cells and Hassall corpuscles

30. *2026 lymphoid lab deck*
The arrowheads point towards which thymic structure?



A. Thymocytes
B. Lymphoblasts
C. Hassall corpuscles
D. Epithelial reticular cells

Answer: D. Epithelial reticular cells

The arrowheads mark **large pale cells** with **oval euchromatic nuclei and nucleoli** among small dark thymocytes = **epithelial reticular cells** (thymic epithelial cells).
A (thymocytes): the many small cells with dense dark nuclei.
B (lymphoblasts): dividing lymphocytes with dark cytoplasm, not pale nuclei with a single nucleolus.
C (Hassall corpuscles): pink concentric whorls in the medulla.

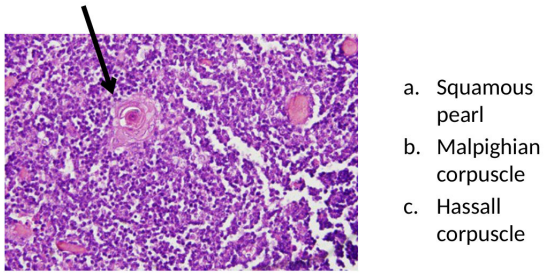
31. *2026 TA review*
A thymic epithelial cell contributes to a barrier that prevents unregulated exposure of developing thymocytes to circulating antigens. Which combination correctly identifies the components of this barrier?

A. Capillary wall, perivascular connective tissue, and thymic epithelial cells
B. Capillary wall, follicular dendritic cells, and reticular fibers
C. High endothelial venules, macrophages, and medullary cords
D. Reticular fibers, endothelial stave cells, and plasma cells

Answer: A. Capillary wall, perivascular connective tissue, and thymic epithelial cells

The **blood-thymus barrier** (cortex) = **capillary wall** (endothelium, basal lamina, pericytes) + **perivascular connective tissue** (with macrophages) + **thymic epithelial cells**.
B: follicular dendritic cells belong to B-cell follicles; the thymus has no reticular fibers.
C: HEVs and medullary cords are lymph-node structures.
D: stave cells and reticular fibers belong to splenic red pulp.

32. *2026 lymphoid lab deck*
Name the indicated thymic structure



A. Squamous pearl
B. Malpighian corpuscle
C. Hassall corpuscle

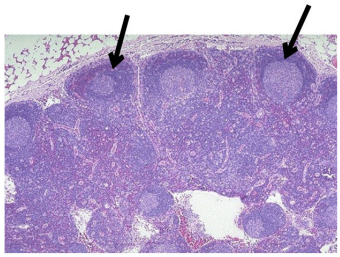
Answer: C. Hassall corpuscle

A pink, **concentric, layered whorl** of flattened epithelial cells in the pale thymic medulla = **Hassall corpuscle**.
A (squamous pearl): a keratin whorl in squamous cell carcinoma, not normal thymus.
B (Malpighian corpuscle): a splenic lymphoid nodule (or a renal corpuscle).

Lymph node organization and identification

33. *2026 lymphoid lab deck*
What lymph node region is identified by the arrows?

Q6- What lymph node region is identified by the arrows?



a. Cortex
b. Paracortex
c. Medulla

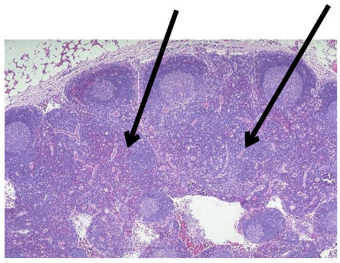
A. Cortex
B. Paracortex
C. Medulla

Answer: A. Cortex

The arrows point at the **round follicles just under the capsule**. Follicles live in the **outer cortex** (B-cell zone).
B (paracortex): lies deeper, under the follicles, and has no follicles.
C (medulla): is the deepest part near the hilum, with cords and pale sinuses.

34. *2026 lymphoid lab deck*
What lymph node region is identified by the arrows?

Q7- What lymph node region is identified by the arrows?



a. Cortex
b. Paracortex
c. Medulla

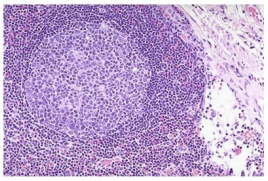
A. Cortex
B. Paracortex
C. Medulla

Answer: B. Paracortex

The arrows reach **past the follicles** into the diffuse, follicle-free tissue above the medulla. That is the **paracortex** (deep cortex, T-cell zone).
A (cortex): the arrows would stop on the round follicles under the capsule.
C (medulla): would show dark cords with pale sinuses near the hilum.

35. *2026 lymphoid session*
Which of the following is present in the germinal center?

Q3 - Which of the following is present in the germinal center?



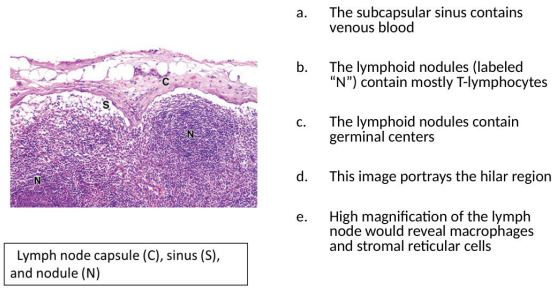
a. Collections of M-cells
b. Aggregate of thymocytes
c. Proliferating lymphocytes
d. Periarteriolar lymphoid sheath
e. Hassall corpuscles

A. Collections of M-cells
B. Aggregate of thymocytes
C. Proliferating lymphocytes
D. Periarteriolar lymphoid sheath
E. Hassall corpuscles

Answer: C. Proliferating lymphocytes

The **pale germinal center** is full of **proliferating B lymphocytes** (activated, dividing, large open nuclei).
A (M cells): epithelial cells over Peyer's patches.
B (thymocytes): developing T cells in the **thymus**.
D (PALS = periarteriolar lymphoid sheath): T-cell cuff in **spleen** white pulp.
E (Hassall corpuscles): pink whorls in the **thymic medulla**.

36. *2026 lymphoid session*
Which of the following is true of the lymph node in this picture?

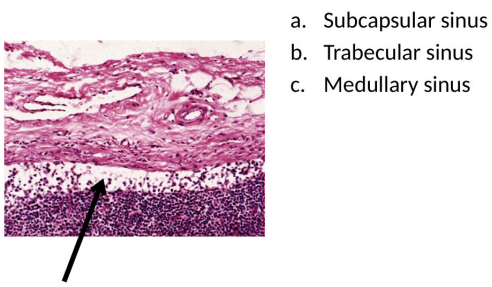


A. The subcapsular sinus contains venous blood
B. The lymphoid nodules (labeled "N") contain mostly T-lymphocytes
C. The lymphoid nodules contain germinal centers
D. This image portrays the hilar region
E. High magnification of the lymph node would reveal macrophages and stromal reticular cells

Answer: E. High magnification of the lymph node would reveal macrophages and stromal reticular cells

The sinus **S** under capsule **C** holds lymph with **macrophages** and a mesh of **reticular cells**, seen at high power.
A (venous blood): sinuses carry **lymph**, not blood.
B (T cells in N): nodules are **B**-cell zones.
C (germinal centers): these nodules are evenly dark, so **primary**.
D (hilum): a convex capsule with a subcapsular sinus is the **cortex**; the hilum is the concave notch.

37. *2026 lymphoid lab deck*
What lymph node sinus is indicated by the arrow?



A. Subcapsular sinus
B. Trabecular sinus
C. Medullary sinus

Answer: A. Subcapsular sinus

The arrow points at the **pale space right under the capsule**, above the dense cortex: the **subcapsular sinus**.
B (trabecular sinus): runs inward along a trabecula through the cortex.
C (medullary sinus): pale channels between cords deep in the medulla.

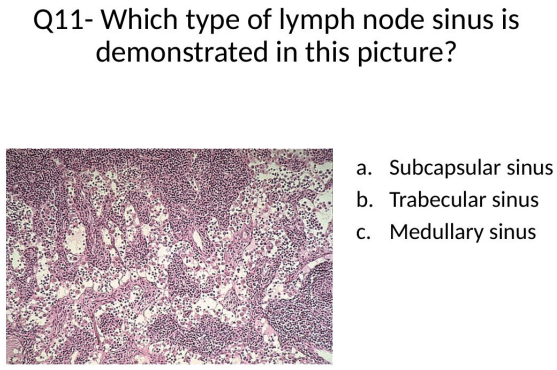
38. *2026 TA review*
Which sequence correctly traces the pathway of lymph flowing through a lymph node?

A. Efferent vessel -> Medullary sinus -> Paracortex -> Subcapsular sinus -> Afferent vessel
B. Subcapsular sinus -> Cortical (trabecular) sinuses -> Medullary sinuses -> Efferent lymphatic vessel
C. High endothelial venule -> Germinal center -> Medullary cord -> Trabecular vein
D. Medullary sinus -> Subcapsular sinus -> Cortical sinus -> Hilum

Answer: B. Subcapsular sinus -> Cortical (trabecular) sinuses -> Medullary sinuses -> Efferent lymphatic vessel

Lymph runs **afferent vessel → subcapsular sinus → cortical (trabecular) sinuses → medullary sinuses → efferent vessel** at the hilum.
A: the same path run backwards.
C: HEVs (high endothelial venules) carry blood, not lymph.
D: starts in the medulla; lymph enters at the capsule.

39. *2026 lymphoid lab deck*
Which type of lymph node sinus is demonstrated in this picture?



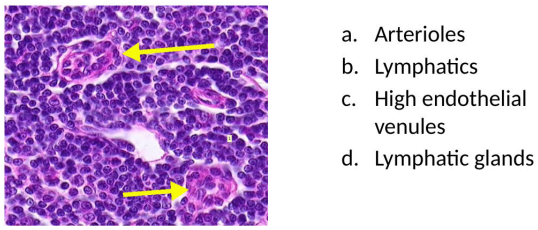
A. Subcapsular sinus
B. Trabecular sinus
C. Medullary sinus

Answer: C. Medullary sinus

The pale channels lie **between dark branching cords**, so they are **medullary sinuses**.
A (subcapsular sinus): one thin pale band right under the capsule.
B (trabecular sinus): runs beside a pink trabecula through the cortex.

Lymph node histology, HEVs, plasma cells

40. *2026 lymphoid lab deck*
What structures are demonstrated by arrows in this high magnification photograph of lymph node paracortex

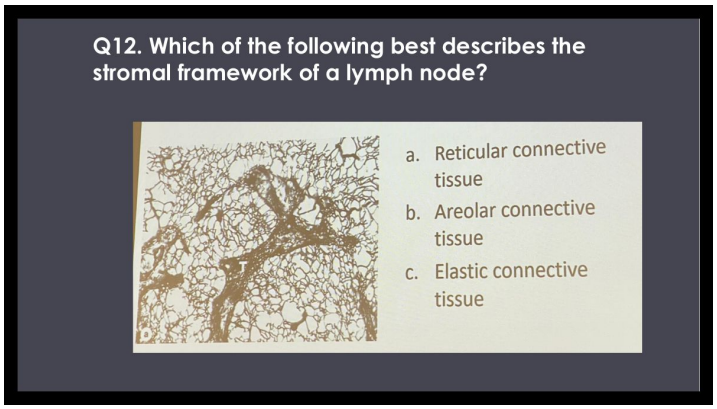


A. Arterioles
B. Lymphatics
C. High endothelial venules
D. Lymphatic glands

Answer: C. High endothelial venules

The yellow arrows mark small vessels in the **paracortex** lined by **plump cuboidal endothelium**: **high endothelial venules (HEVs)**.
A (arterioles): have a smooth muscle wall and flat endothelium.
B (lymphatics): very thin walls, flat endothelium.
D (lymphatic glands): not a real structure.

41. *2026 lymphoid lab deck*
Which of the following best describes the stromal framework of a lymph node?



A. Reticular connective tissue
B. Areolar connective tissue
C. Elastic connective tissue

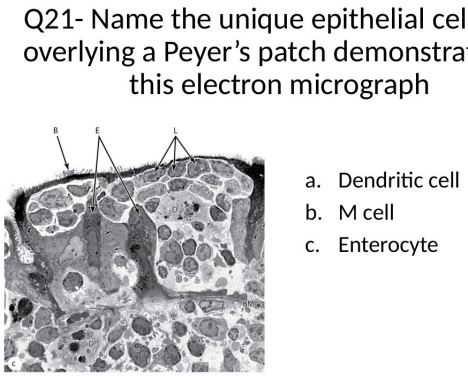
Answer: A. Reticular connective tissue

Silver stain turns **reticular fibers** (type III collagen) **black**. The fine black mesh is **reticular connective tissue**, the stroma of lymph nodes (and spleen and marrow). T marks a denser trabecula.
B (areolar CT = connective tissue): loose mix of fibers under epithelia, not a fine mesh.
C (elastic CT): in large arteries and ligaments; needs an elastic stain.

MALT: tonsils, Peyer's patches, appendix (incl. IDs)

42. *2026 lymphoid lab deck*

Name the unique epithelial cell type overlying a Peyer's patch demonstrated in this electron micrograph



A. Dendritic cell

B. M cell

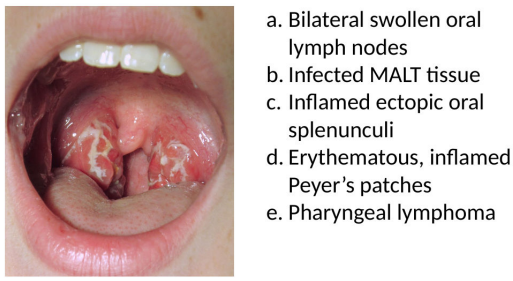
C. Enterocyte

Answer: B. M cell

The cells capping the pale pockets full of lymphocytes (**L**) and dendritic cells (**D**) are **M (microfold) cells**. They have short folds, **no brush border**, and a basal pocket.
C (enterocytes, E): the neighboring cells with a **brush border** (B).
A (dendritic cells, D): sit inside the pocket and below; they are not the epithelial cell asked about.

43. *2026 lymphoid session*

A 10-year-old boy presented to his doctor with fever, sore throat, and tender, swollen cervical lymph nodes. The doctor examined the oral cavity and prescribed antibiotics. Which of the following best applies to the abnormalities visible on examination of the oral cavity?



A. Bilateral swollen oral lymph nodes

B. Infected MALT tissue

C. Inflamed ectopic oral splenunculi

D. Erythematous, inflamed Peyer's patches

E. Pharyngeal lymphoma

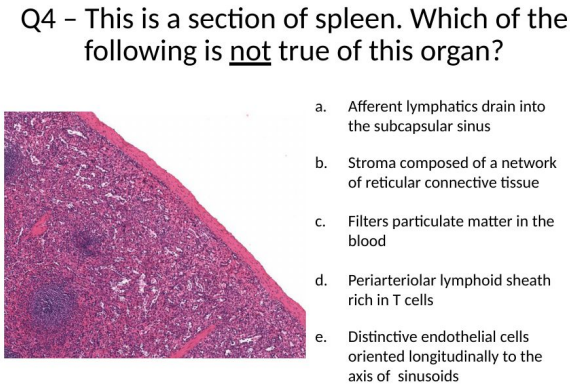
Answer: B. Infected MALT tissue

Sore throat, fever, tender neck nodes and **red tonsils with white exudate** = **tonsillitis**. The palatine tonsils are **MALT** (mucosa-associated lymphoid tissue), so this is infected MALT.
A: there are no lymph nodes in the oral mucosa; the swollen nodes are in the neck.
C: splenunculi (accessory spleens) sit near the spleen, not in the mouth.
D: Peyer's patches are in the **ileum**.
E: acute fever with exudate in a child points to infection, not lymphoma.

Spleen structure: white and red pulp (incl. IDs)

44. *2025 faculty lymphoid session*

This is a section of spleen. Which of the following is not true of this organ?



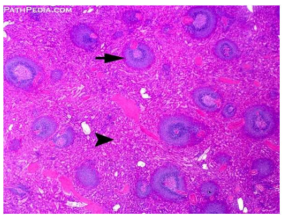
A. Afferent lymphatics drain into the subcapsular sinus
B. Stroma composed of a network of reticular connective tissue
C. Filters particulate matter in the blood
D. Periarteriolar lymphoid sheath rich in T cells
E. Distinctive endothelial cells oriented longitudinally to the axis of sinusoids

Answer: A. Afferent lymphatics drain into the subcapsular sinus

A is false: the spleen filters **blood**, not lymph. It has **no afferent lymphatics** and **no subcapsular sinus**; red pulp sits right under the thick capsule. Afferent lymphatics into a subcapsular sinus = **lymph node**.
B (reticular stroma): true; reticular fibers and cells form the framework.
C (filters blood): true; it removes old RBCs (red blood cells) and particles.
D (PALS = periarteriolar lymphoid sheath): true; T-cell rich.
E: true; **stave cells** run lengthwise along the sinusoids.

45. *2026 lymphoid lab deck*

The splenic structure demonstrated by the arrow is…



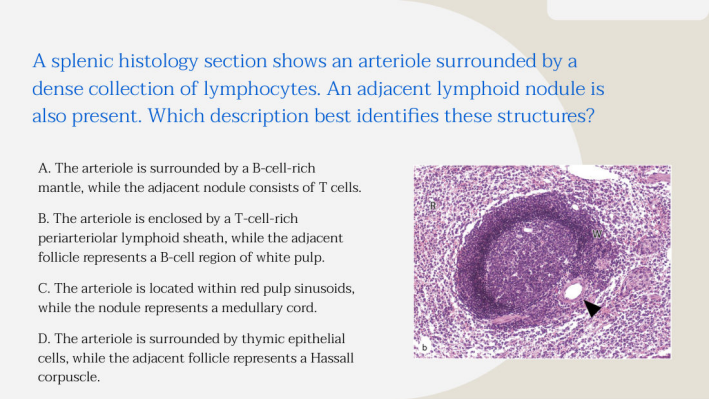
A. White pulp
B. Red pulp
C. White matter
D. Gray matter
E. Trabeculum

Answer: A. White pulp

The arrow sits on a round **dark purple** lymphoid nodule with a central arteriole = **white pulp**.
B (red pulp): the pink tissue between the nodules (the arrowhead).
E (trabeculum): the pale pink, smooth, acellular connective tissue bands carrying vessels.
C / D (white/gray matter): nervous system terms, not spleen.

46. *2026 TA review*

A splenic histology section shows an arteriole surrounded by a dense collection of lymphocytes. An adjacent lymphoid nodule is also present. Which description best identifies these structures?



A. The arteriole is surrounded by a B-cell-rich mantle, while the adjacent nodule consists of T cells.

B. The arteriole is enclosed by a T-cell-rich periarteriolar lymphoid sheath, while the adjacent follicle represents a B-cell region of white pulp.

C. The arteriole is located within red pulp sinusoids, while the nodule represents a medullary cord.

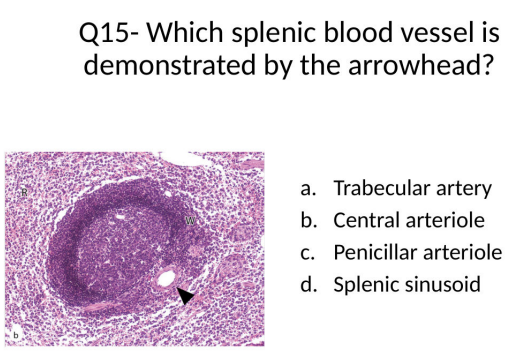
D. The arteriole is surrounded by thymic epithelial cells, while the adjacent follicle represents a Hassall corpuscle.

Answer: B. The arteriole is enclosed by a T-cell-rich periarteriolar lymphoid sheath, while the adjacent follicle represents a B-cell region of white pulp.

The arrowhead marks the **central arteriole**, cuffed by the T-cell **PALS** (periarteriolar lymphoid sheath); the big round **nodule** beside it is the **B-cell** follicle of white pulp.
A: swaps T and B.
C: medullary cords belong to a **lymph node**; this is spleen white pulp.
D: thymic epithelial cells and Hassall corpuscles are **thymus** only.

47. *2026 lymphoid lab deck*

Which splenic blood vessel is demonstrated by the arrowhead?



A. Trabecular artery

B. Central arteriole

C. Penicillar arteriole

D. Splenic sinusoid

Answer: B. Central arteriole

The arrowhead is a small **muscular arteriole inside white pulp**, at the edge of a nodule = **central arteriole**.
A (trabecular artery): larger, inside a pink connective tissue trabecula.
C (penicillar arteriole): small straight branches out in **red pulp**, after the central arteriole leaves white pulp.
D (sinusoid): thin-walled, wide, lined by stave cells, in red pulp.

Splenic blood flow and RBC removal

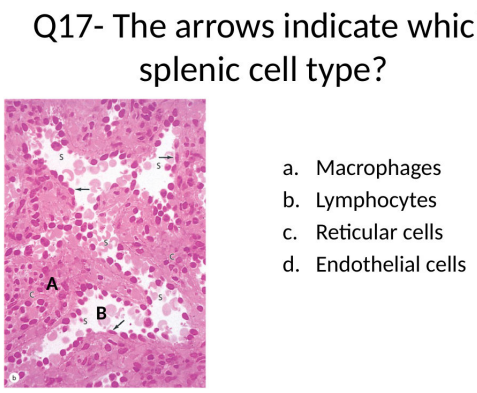
48. *2026 TA review*
In the open theory of splenic circulation, where is blood discharged from the penicillar arterioles?

A. Directly into the lumen of the splenic vein
B. Into the central arteriole of the white pulp
C. Into the subcapsular sinus of the spleen
D. Directly into the meshwork/stroma of the splenic cords (cords of Billroth)

Answer: D. Directly into the meshwork/stroma of the splenic cords (cords of Billroth)

In **open** circulation the penicillar arterioles/capillaries end **open**, dumping blood straight into the **splenic cords** (cords of Billroth) of red pulp.
Cells must then squeeze between stave cells back into sinusoids; old rigid RBCs (red blood cells) cannot and are eaten by cord macrophages.
A (splenic vein) / E (trabecular vein): come last, after the sinusoids.
Sinusoids directly: that is **closed** circulation.
Central arteriole: comes before, in white pulp. **Subcapsular sinus:** spleen has none.

49. *2026 lymphoid lab deck*
The arrows indicate which splenic cell type?



A. Macrophages
B. Lymphocytes
C. Reticular cells
D. Endothelial cells

Answer: D. Endothelial cells

The arrows mark **elongated nuclei** lining the open sinusoid lumens (S) and **bulging into them** = **endothelial cells** of splenic sinusoids, called **stave cells**.
Macrophages: large cells with pale cytoplasm, often full of RBCs (red blood cells), sitting **in the cords** (C), not lining lumens.
Lymphocytes: small round dark nuclei scattered in cords.
Reticular cells: stromal cells inside the cords. **Epithelial reticular cells:** thymus only.

Clinical correlates: metastasis, splenomegaly, integration

50. *2026 lymphoid session*
A 47-year-old woman detected a cancerous lump in the upper outer quadrant of her breast. Her physician recommended surgical removal of the mass with a sentinal lymph node biopsy of the first axillary lymph node that the cancer was most likely to spread to from the primary tumor. If pathologic examination failed to reveal tumor in the sentinal lymph node, is the likelihood of finding positive lymph nodes in the remaining axillary lymph nodes increased or decreased?.

A. Increased likelihood of positive axillary lymph nodes
B. Decreased likelihood of positive axillary lymph nodes

Answer: B. Decreased likelihood of positive axillary lymph nodes

The **sentinel node** is the **first node** to receive lymph from the tumor, so tumor cells reach it first. If it is **negative**, spread to the rest of the axillary nodes is **less likely**.
A (increased): the trap; a clean sentinel node lowers, not raises, the chance of other positive nodes.